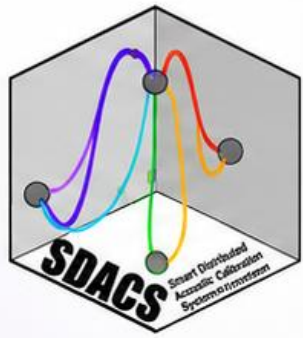


# Smart Distributed Acoustic Calibration System (SDACS)

Capstone II - Electronic Systems Engineering

Conestoga College

August 12, 2026



# AGENDA

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INTRODUCTION

2

SYSTEM REQUIREMENTS

3

SYSTEM DESIGN

4

TEST CRITERIA

5

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CONCLUSION

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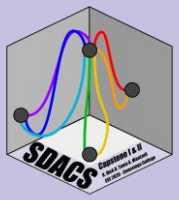
RECOMMENDATIONS

9

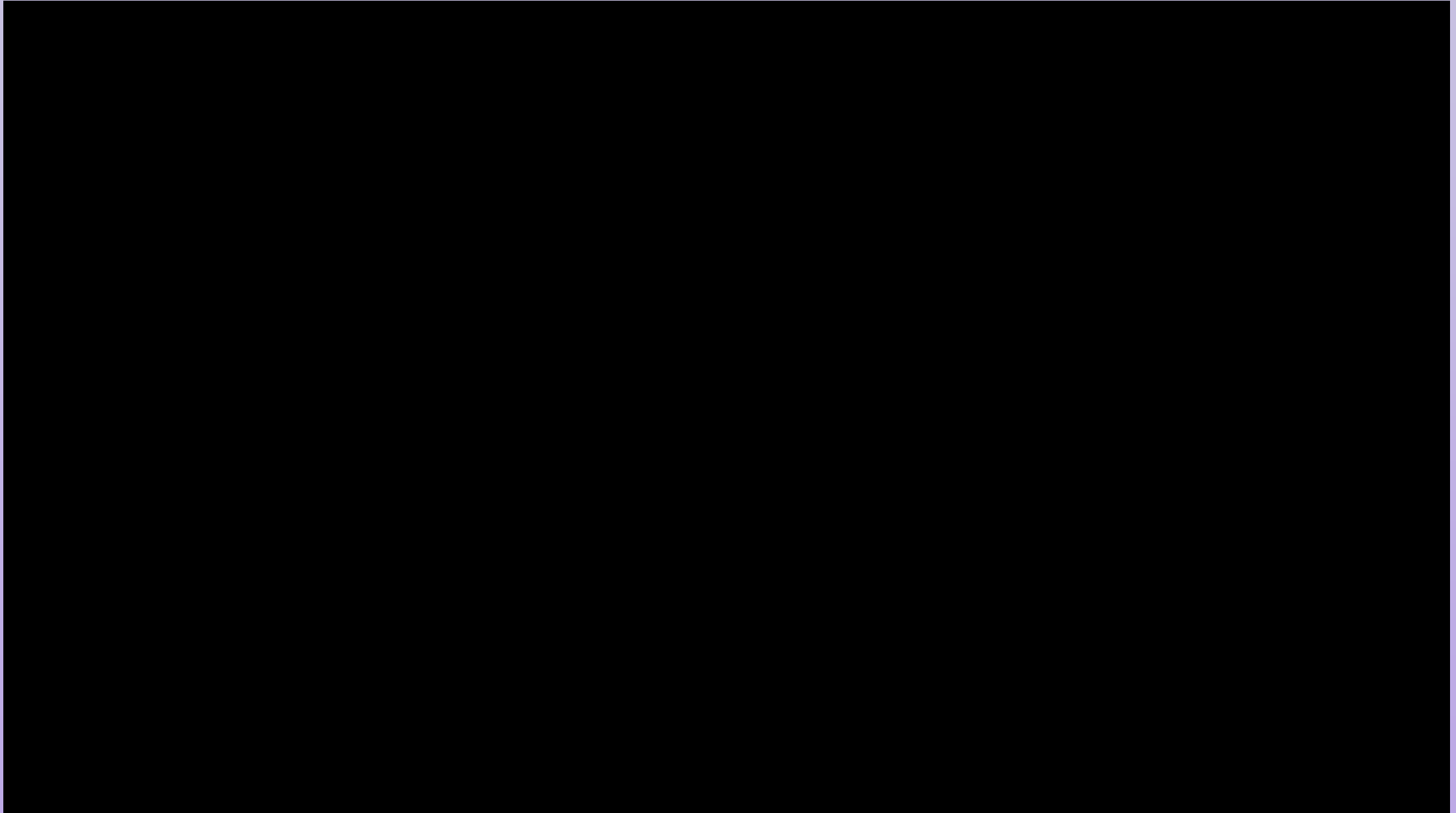
DEMO (8–10 Min)

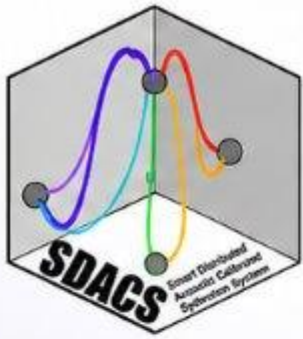
10

Q & A



# 1. INTRODUCTION





## 2. SYSTEM REQUIREMENTS

1

### Multi-Node Embedded Acoustic System



- Minimum of 3 recording nodes
- Server-controlled

2

### Reliable Embedded Firmware Operation

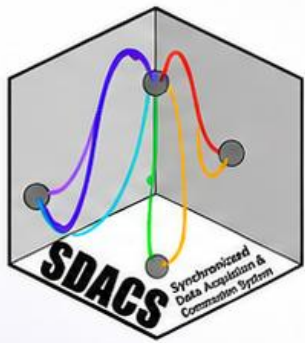


- Chunk-based buffering for audio recording
- Connection-checking and time-outs for Wi-Fi and Bluetooth



These requirements drove the **SDACS** architecture and implementation choices.





## 2. SYSTEM REQUIREMENTS

3



### Requirement 3: Coordinated Multi-Node Recording

- Synchronous recording captures (minimized delay)

4



### Requirement 4: Wireless Data Aggregation and Processing

- Nodes send data results to Server
- FFT, SPL, Battery Life (%)

5

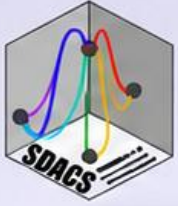


### Requirement 5: Verification, Validation, and Visualization

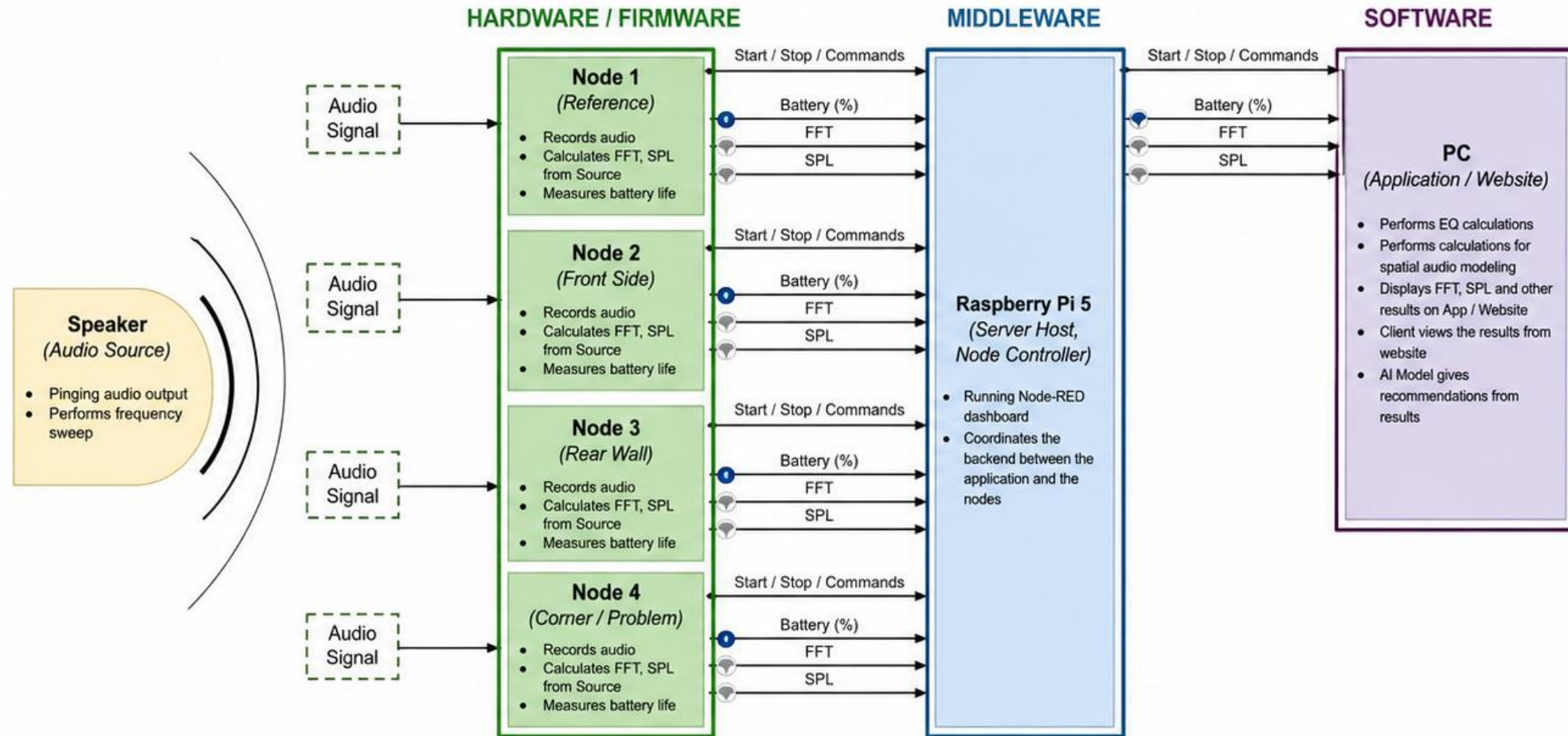
- Dashboard graphing results, analysis and recommendations



These requirements define synchronized capture, wireless reporting, and operator-facing analysis.



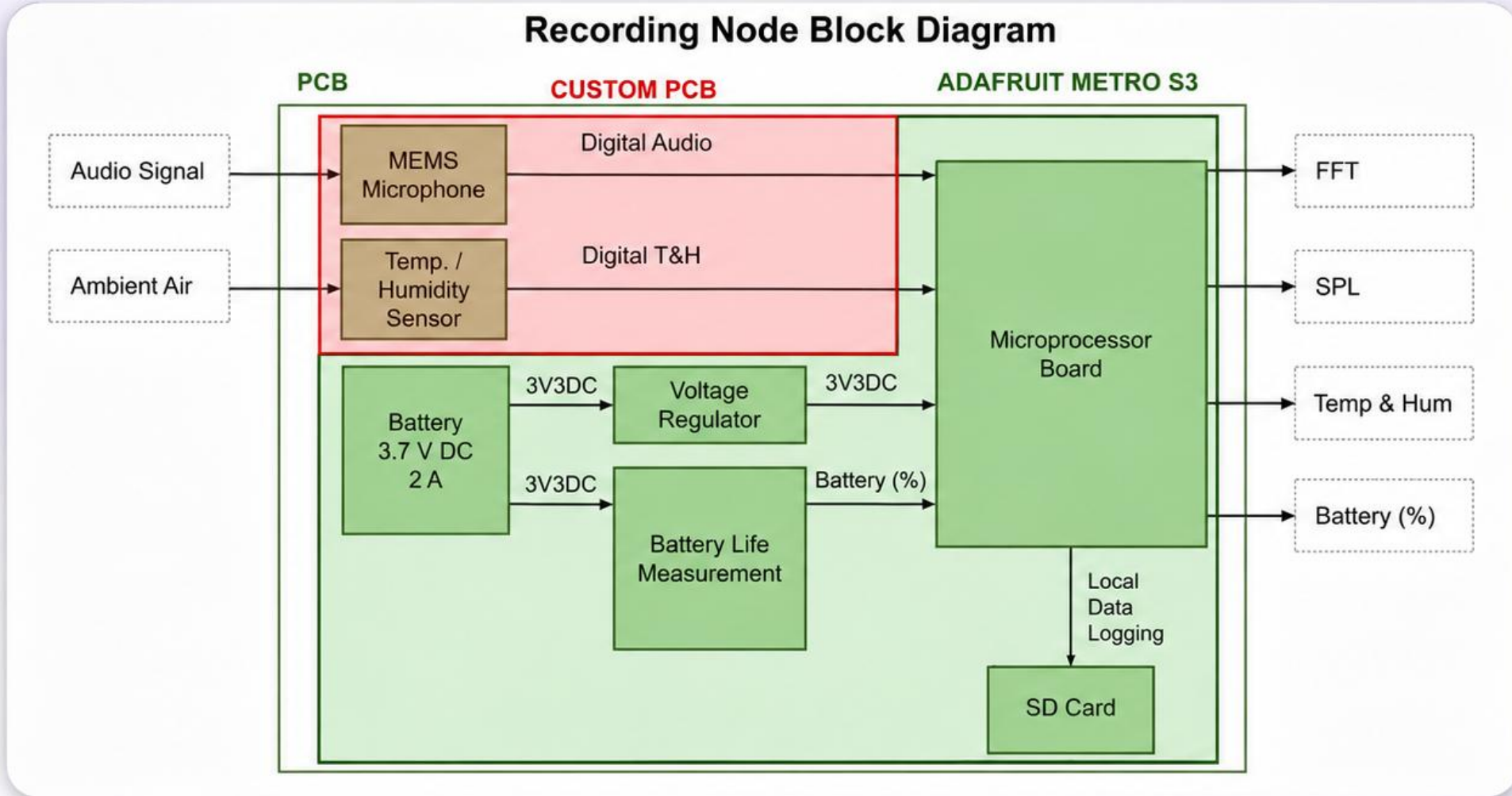
# 3. SYSTEM DESIGN – OVERVIEW



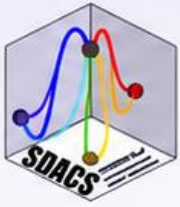
**End-to-end system architecture:** Audio is captured at multiple nodes, processed by the Raspberry Pi 5, and insights are delivered to the PC application for analysis and recommendations.



# 3. SYSTEM DESIGN – HARDWARE

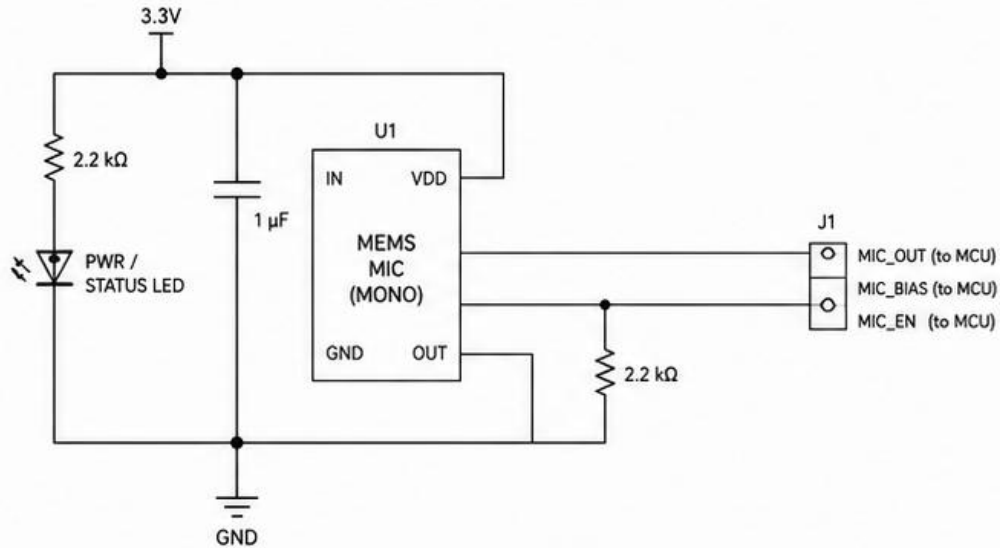


**Integrated sensing, processing, and power management** on a compact recording node for accurate, efficient, and autonomous acoustic monitoring.

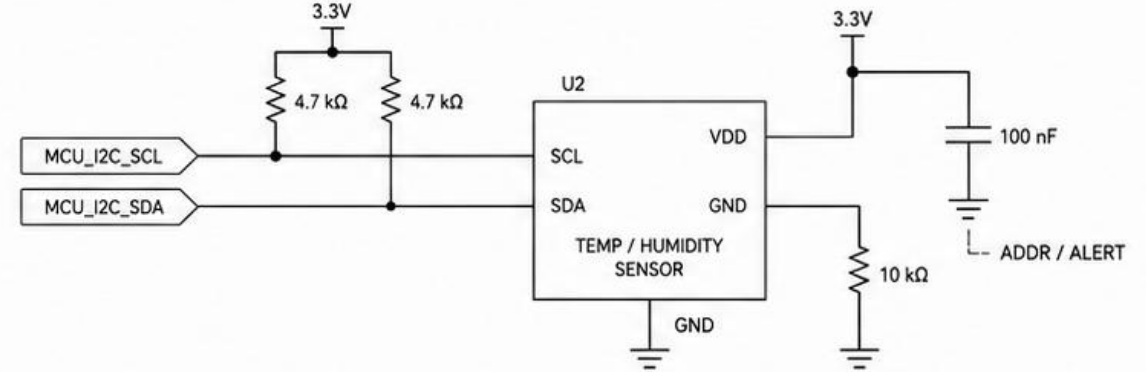


# SYSTEM DESIGN – HARDWARE SCHEMATICS

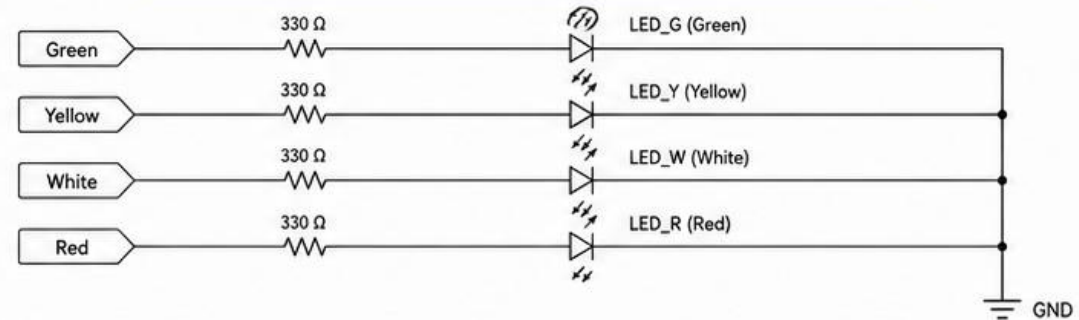
## 1. MEMS Microphone Circuit (Mono)



## 2. Temperature / Humidity Sensor

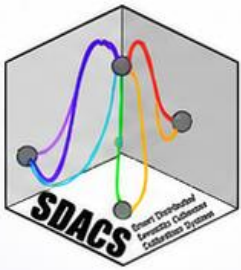


## 3. Power LEDs (Battery Life)



Hardware schematics capture the microphone front end, environmental sensing, and battery-status indication used in the SDACS node design.





# 3. SYSTEM DESIGN - HARDWARE



## Custom PCB



ICS-43432 – MEMS microphone



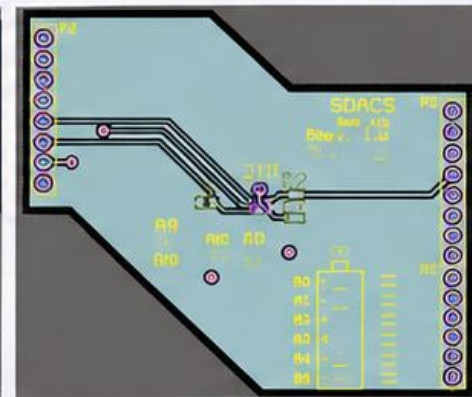
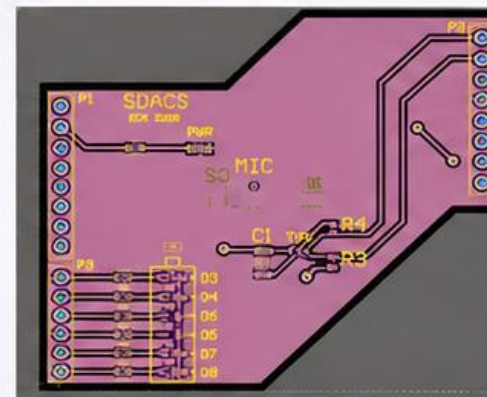
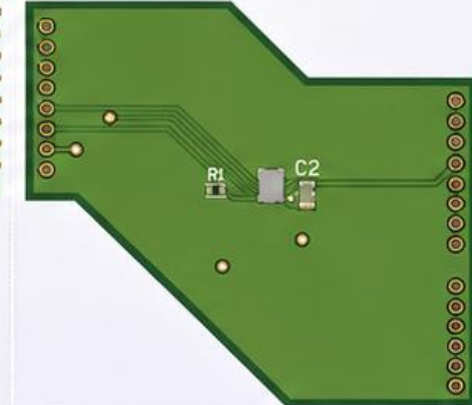
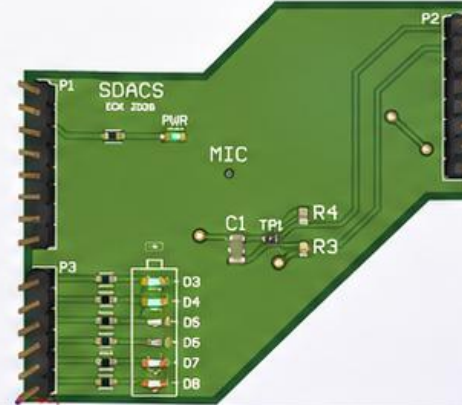
SHT41 – temperature & humidity sensor



LED array – battery life indicator



Custom shape – debug access (BOOT, etc.)



FRONT

BACK

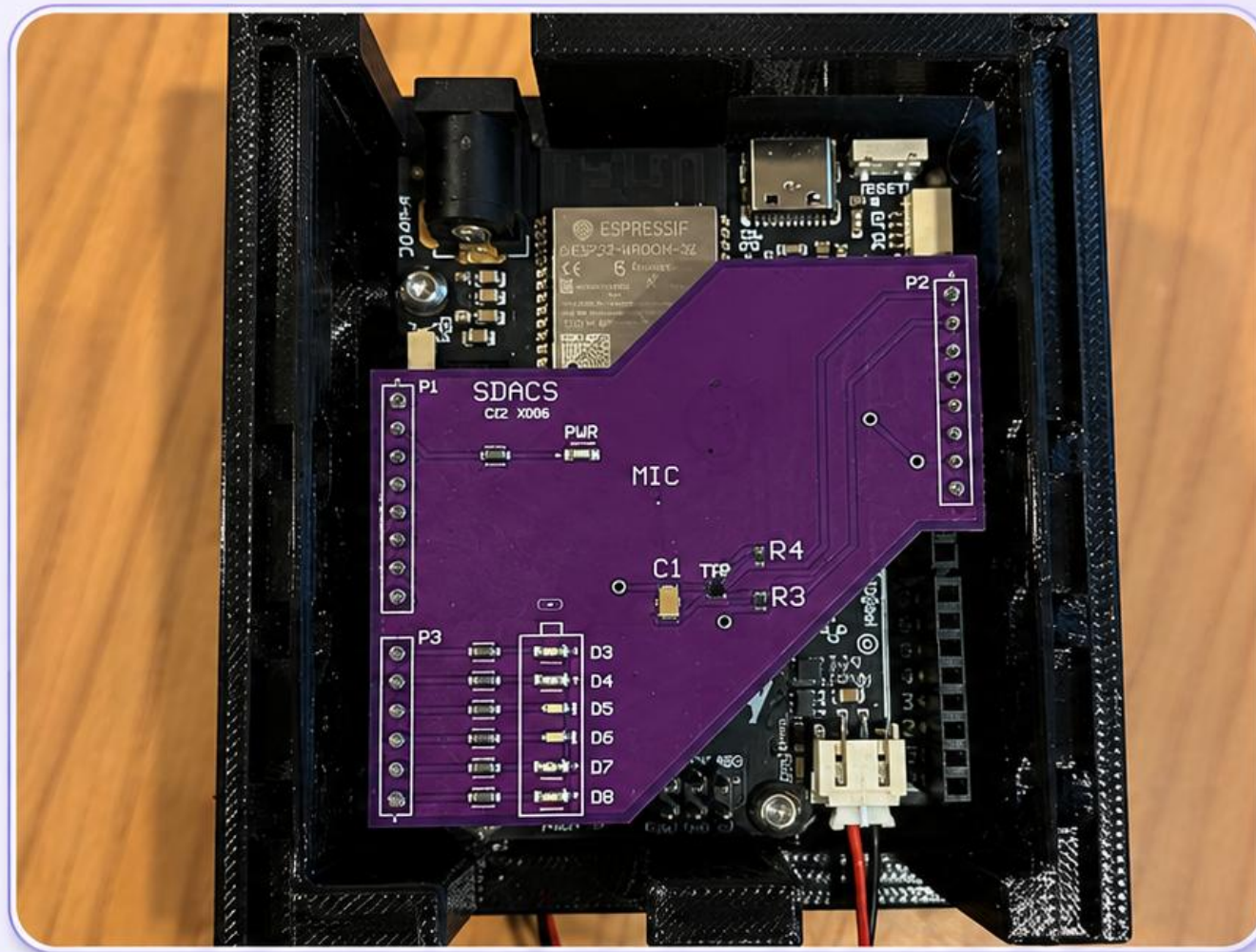


The custom **PCB** integrates sensing, status indication, and service access in a compact, application-specific board.



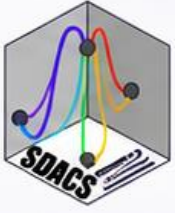


# 3. SYSTEM DESIGN - HARDWARE



Assembled SDACS node hardware integrating the ESP32-based controller, custom PCB, and protective enclosure.



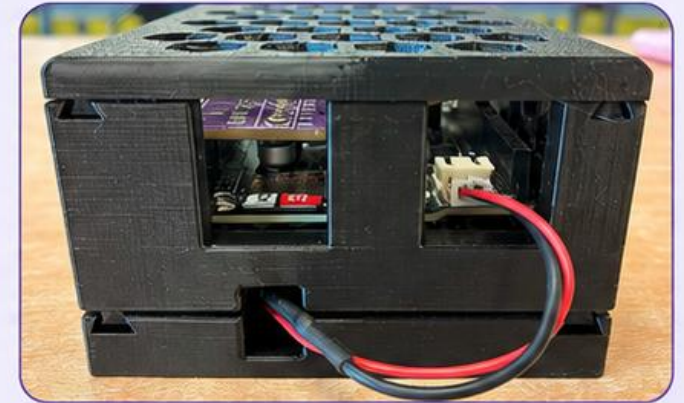
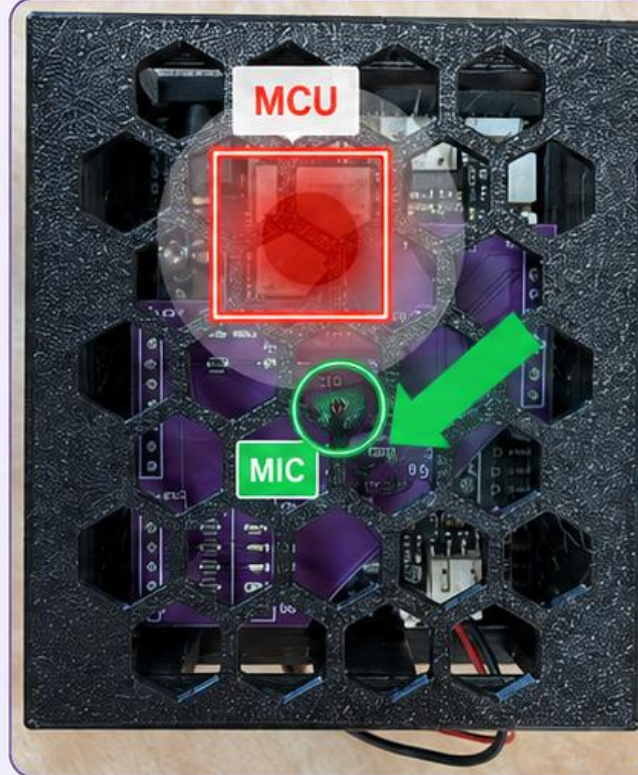


# 3. SYSTEM DESIGN - HARDWARE

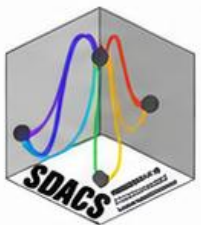


## Key Design Considerations

- 1  SHT41 placement
- 2  ICS-43432 placement
- 3  Access to debug pins (enclosure)

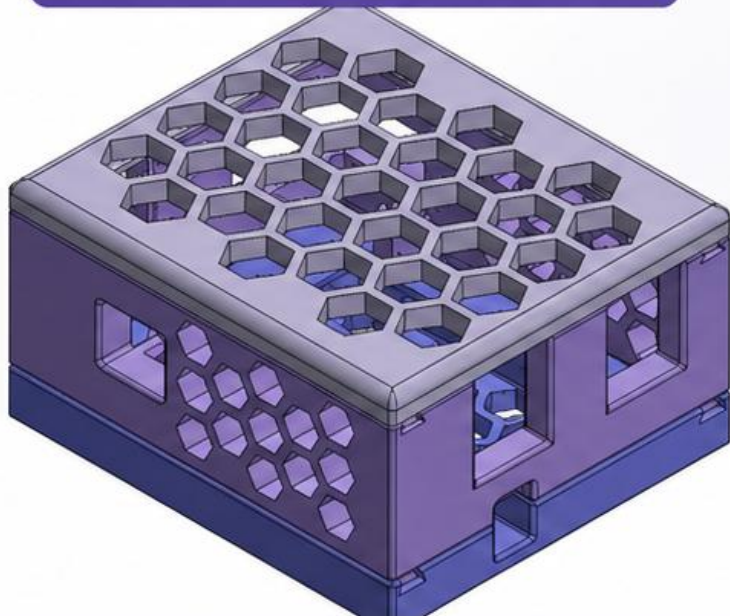






# 3. SYSTEM DESIGN - HARDWARE

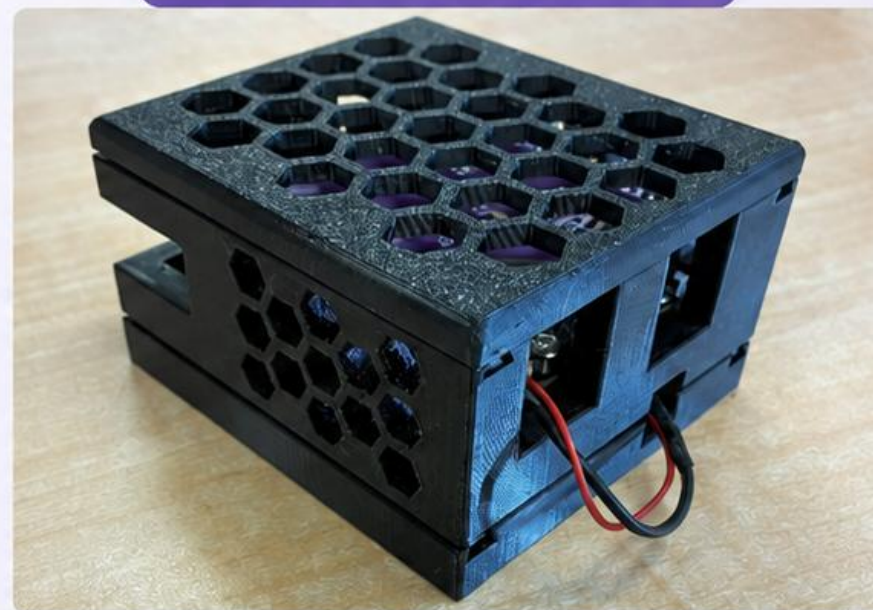
## CAD ENCLOSURE CONCEPT



3D model of the SDACS enclosure optimized for sensors, airflow and access.



## FABRICATED PROTOTYPE



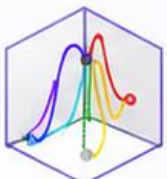
3D printed enclosure assembled with electronics and wiring.



**Successfully translated from concept to working hardware.**

The SDACS enclosure meets functional, integration and manufacturing goals.





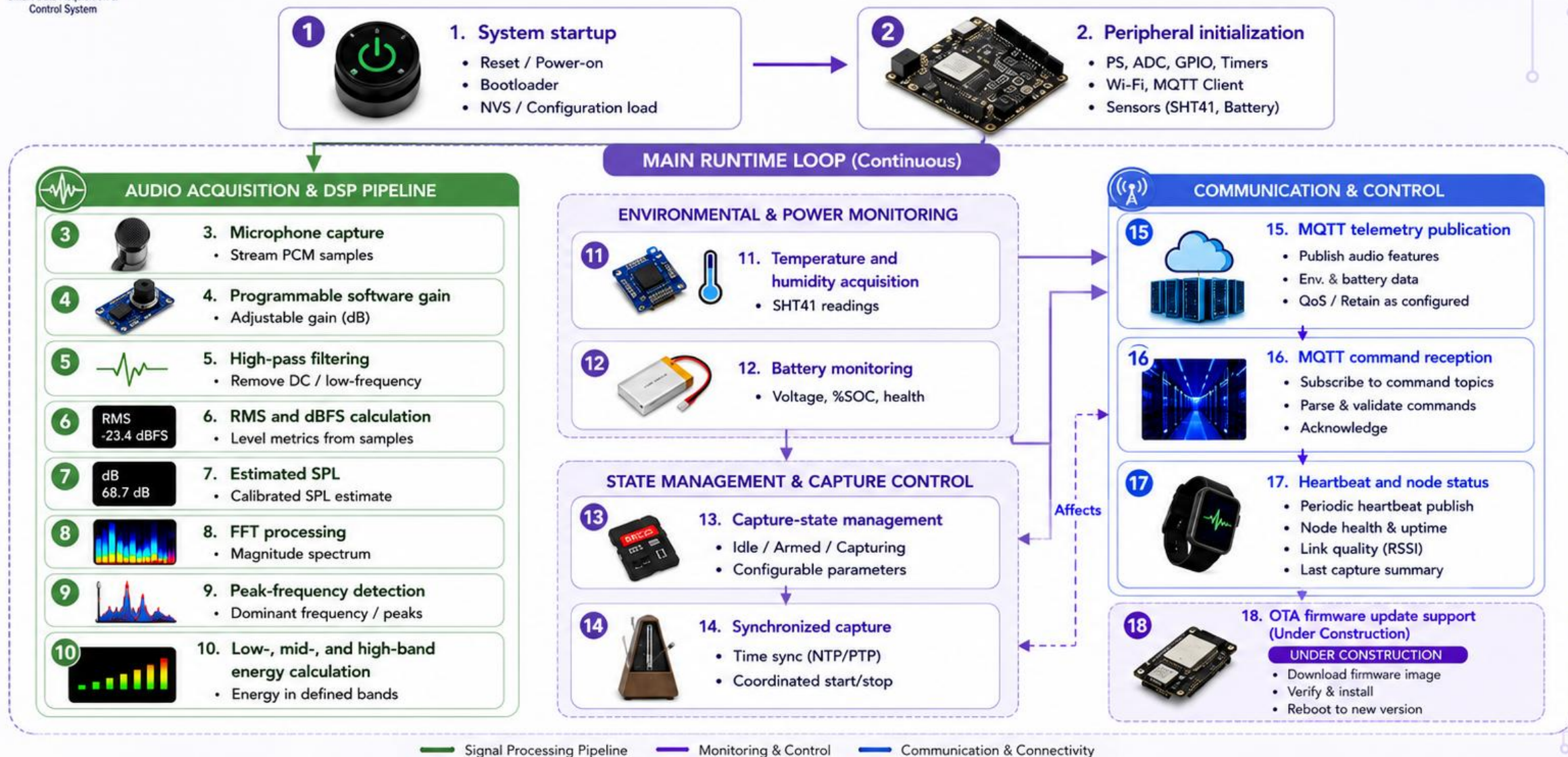
**SDACS**

Smart Data Acquisition &  
Control System

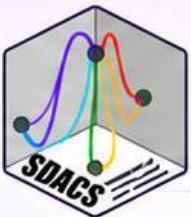
# 3. SYSTEM DESIGN – FIRMWARE

## ESP-IDF Firmware Functional Units Process Flow

*SDACS Sensor Node Runtime and Control Flow*



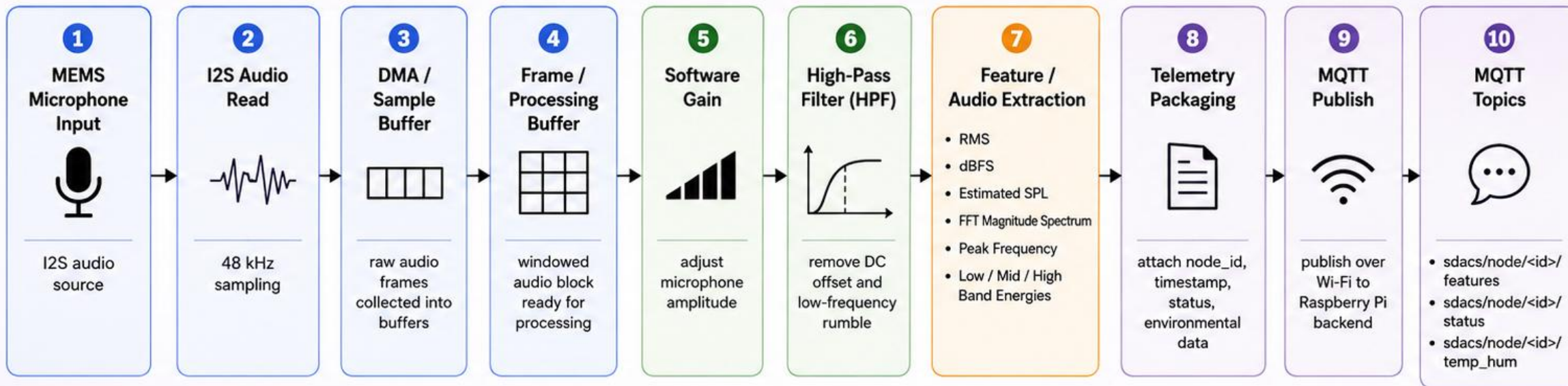




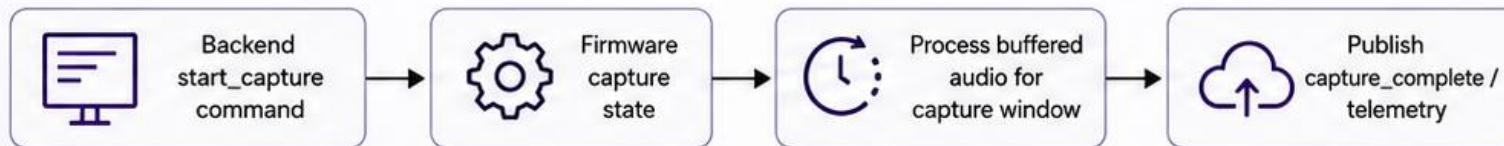
# 3. SYSTEM DESIGN – FIRMWARE

## SDACS Firmware Audio Processing Flow

*HPF and Gain Added to Existing Feature Extraction and MQTT Streaming Pipeline*



### Command / Control Path (Secondary)



### Why HPF and Gain Before Feature Extraction?

Preprocessing improves signal quality before feature computation and MQTT transmission.



Audio Input / Buffering



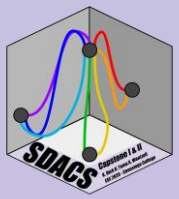
Preprocessing (Gain + HPF)



Feature Extraction

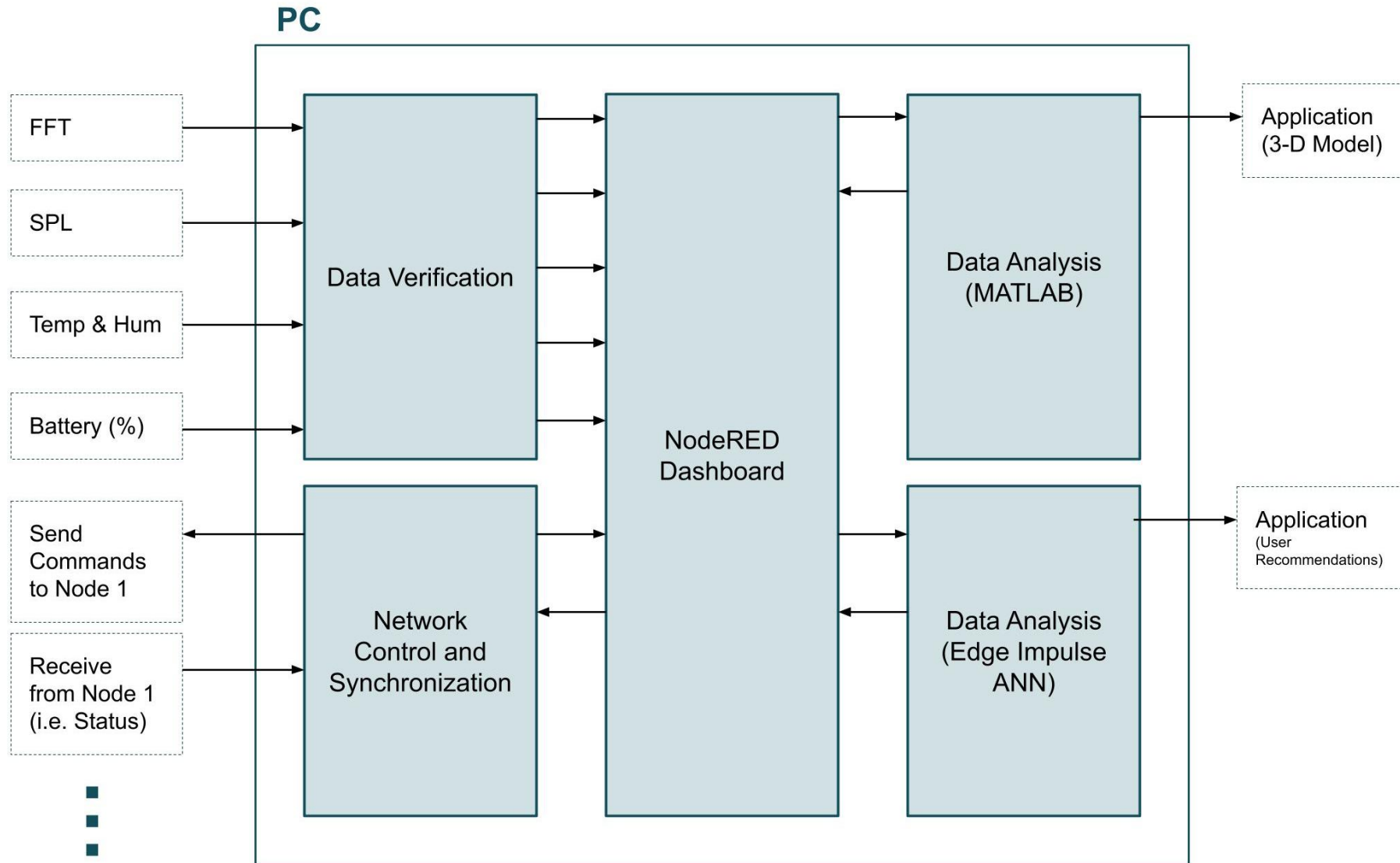


MQTT / Network



# 3. SYSTEM DESIGN – MIDDLEWARE

Server Block Diagram







# 3. SYSTEM DESIGN – MIDDLEWARE

Middleware coordination centered on the node firmware state machine.

## HOW A CAPTURE STARTS



## FIRMWARE STATE MACHINE (PER NODE)

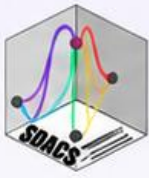


## SYNCHRONIZED FOUR-NODE CAPTURE



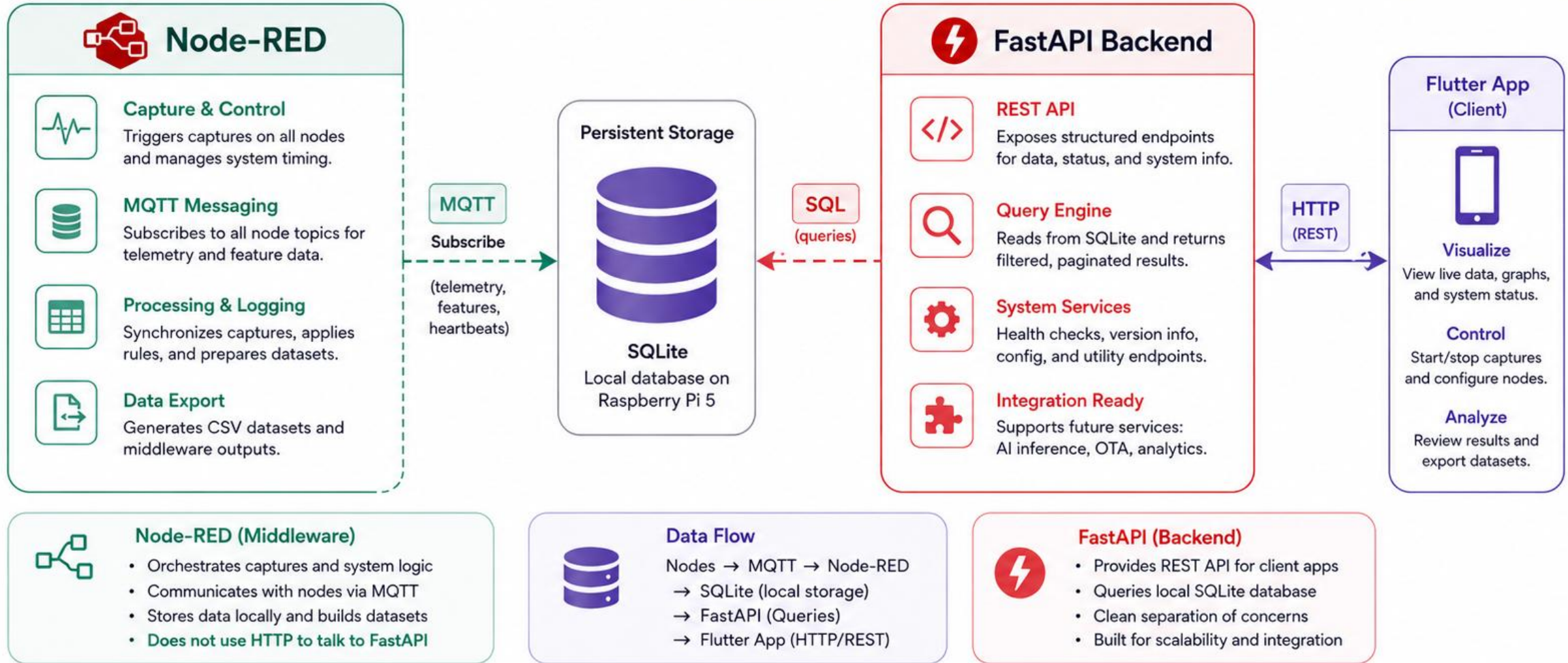
### KEY TAKEAWAY

Middleware coordinates capture commands, while each node follows the same predictable state sequence from idle to result publication.



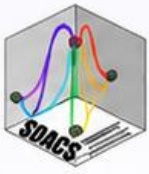
# 3. SYSTEM DESIGN – MIDDLEWARE

Node-RED for control & diagnostics • MQTT for messaging • FastAPI for structured backend services



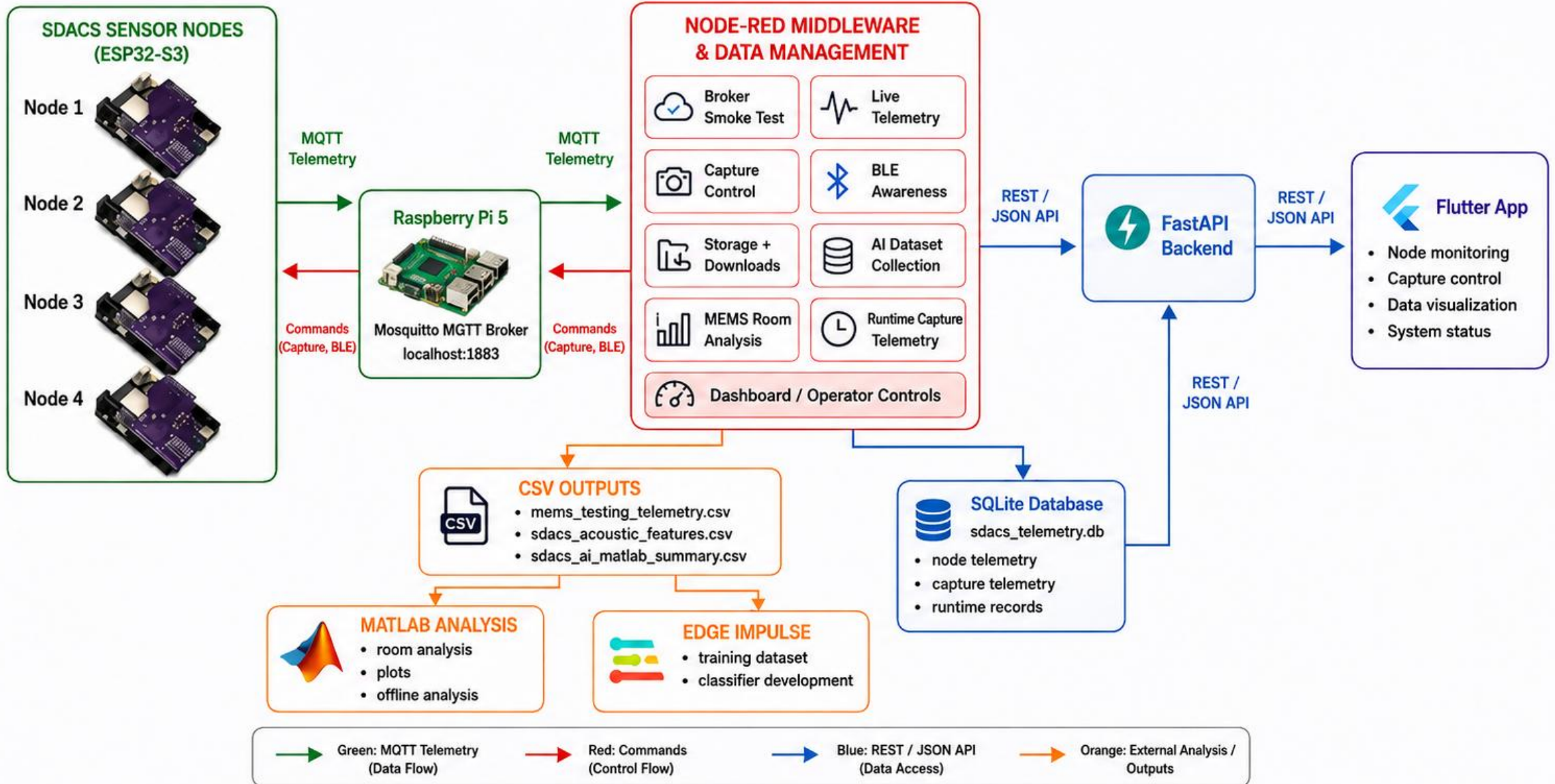
**Separation of Concerns:** Node-RED handles real-time messaging and orchestration via MQTT. FastAPI provides structured, secure access to data via HTTP for client applications.

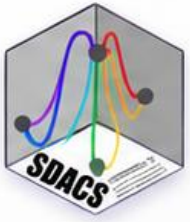




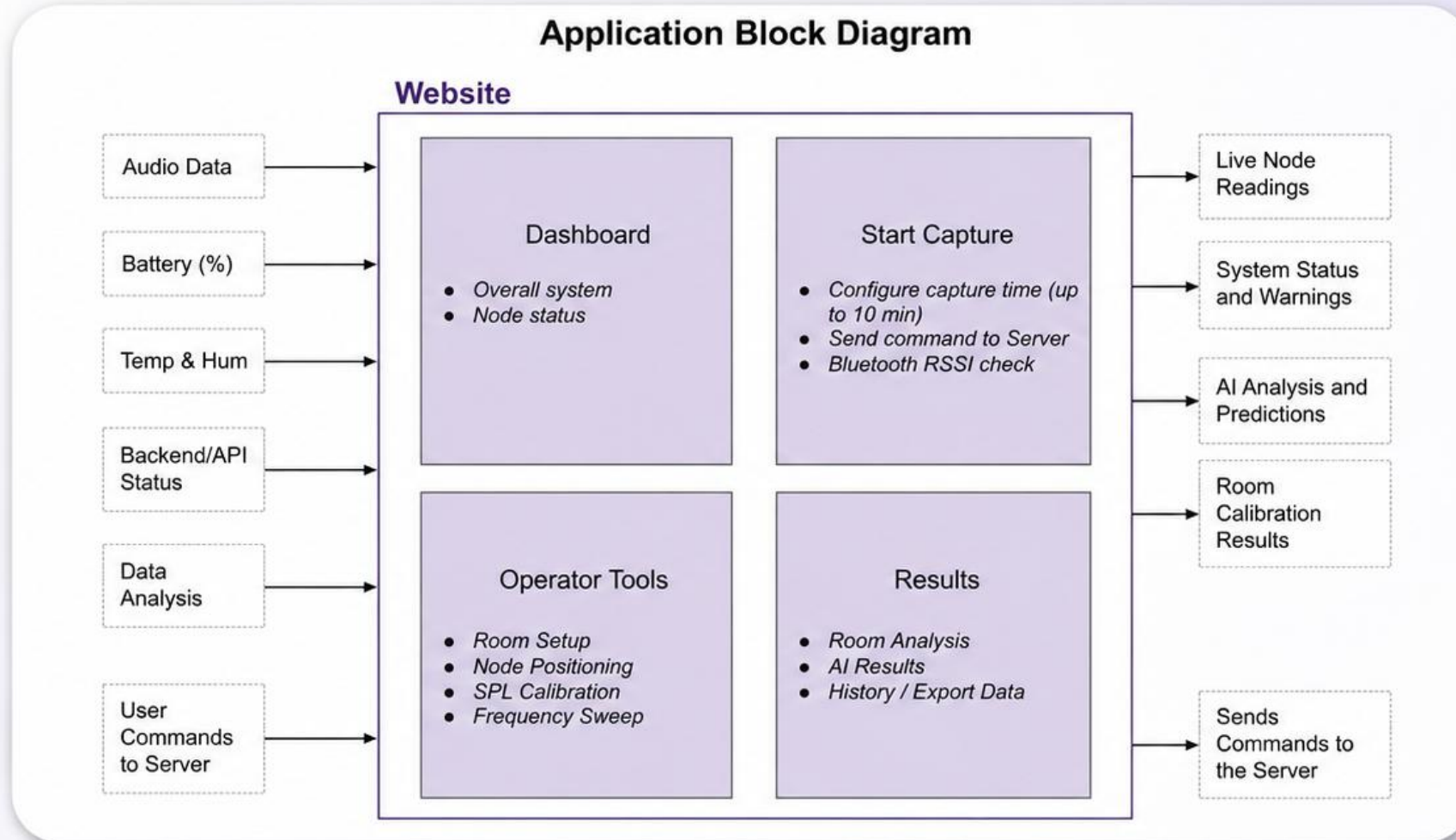
# 3. SYSTEM DESIGN – MW/SW

## SDACS Node-RED, FastAPI, and Flutter Architecture



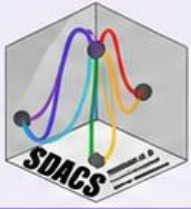


### 3. SYSTEM DESIGN - SOFTWARE



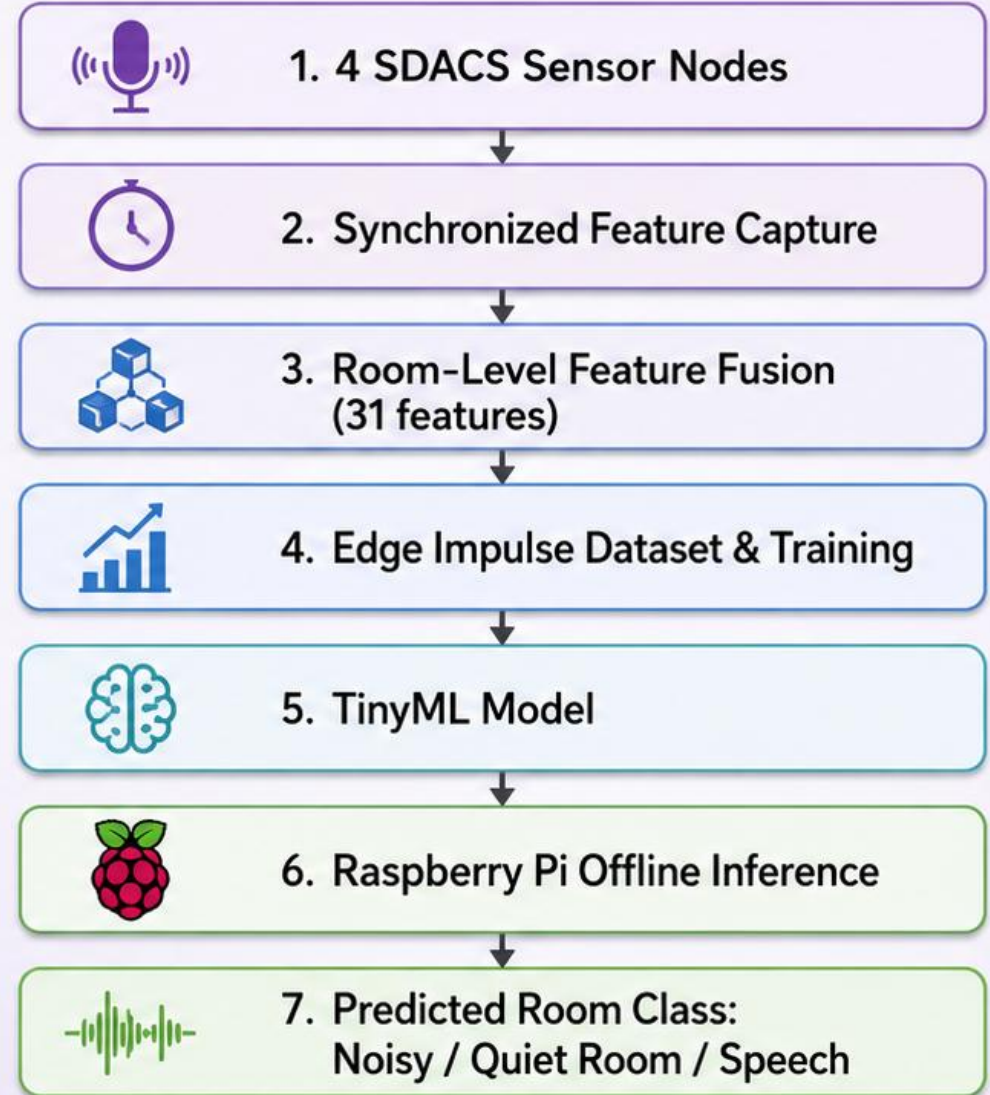
The software layer provides the operator interface for live monitoring, capture control, analysis, and results visualization.

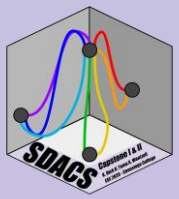




# 3. SYSTEM DESIGN – EDGE IMPULSE

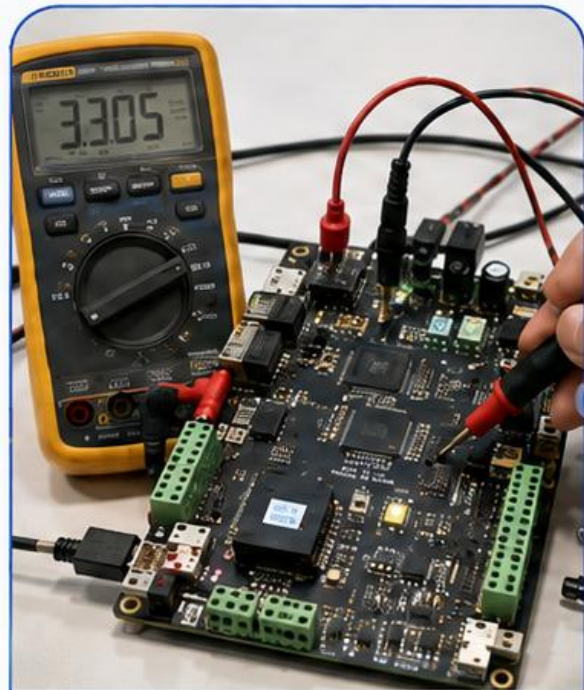
- Develop an intelligent acoustic classification system for the SDACS platform using **TinyML**.
- Classify room environments as **Noisy**, **Quiet Room with White Noise**, or **Speech**.
- Extract and **fuse** acoustic features from synchronized sensor data.
- Enable **fast, low-latency, offline** inference on the Raspberry Pi with a lightweight neural network.
- Support future room **calibration** and intelligent acoustic **optimization**.





# 4. TEST CRITERIA: Verification Strategy

Hardware → Firmware → Middleware → Software



## Hardware

Pinout and power verification



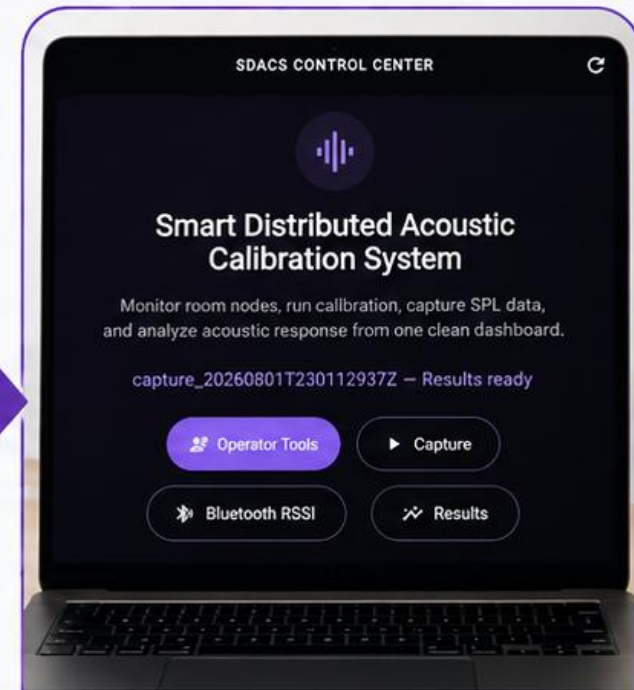
## Firmware

Audio, sensing, and logging



## Middleware

MQTT, API, and storage



## Software

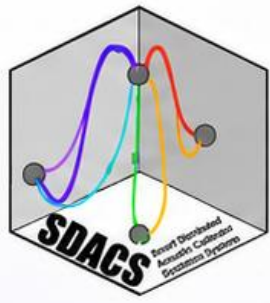
Flutter display and controls



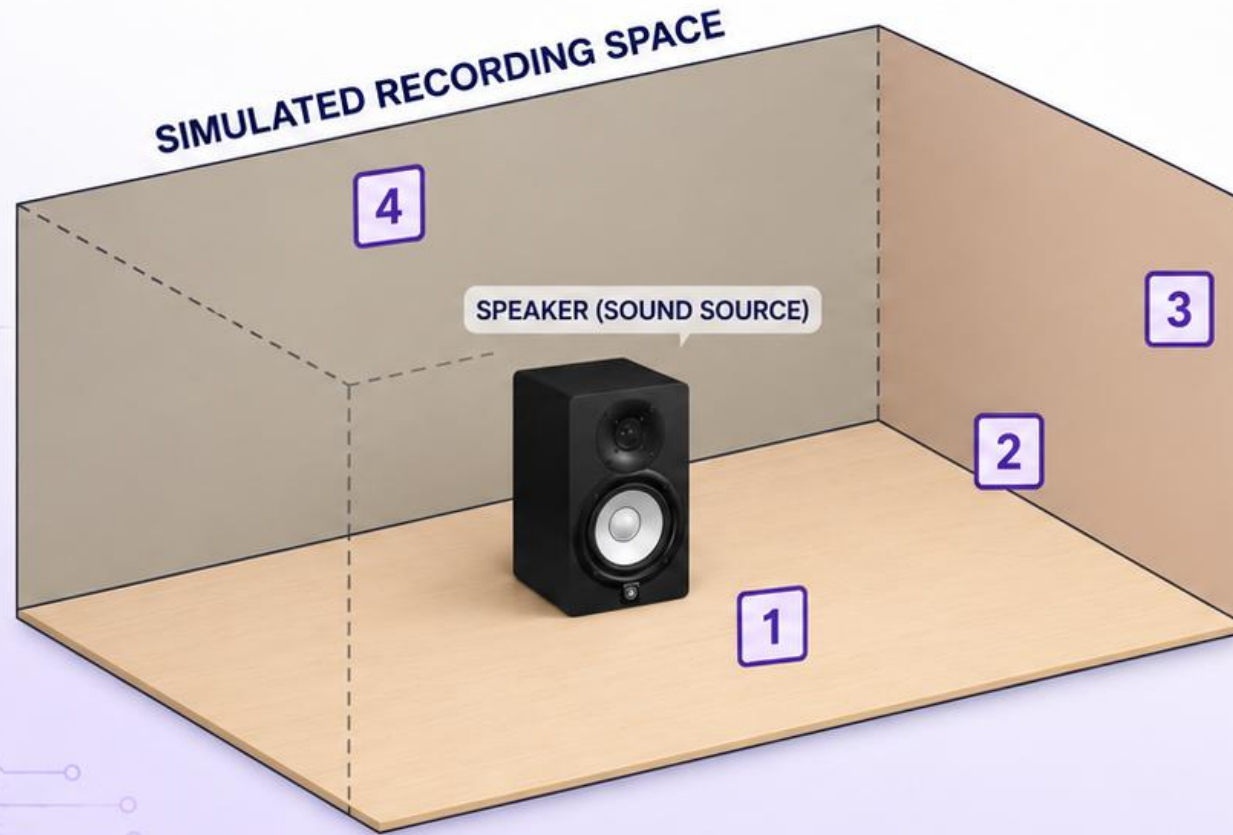
Integrated system validation across all SDACS functional layers







# 5. PROTOTYPING AND TESTING - TEST SETUP



## SPEAKER (SOUND SOURCE)

Yamaha HS5 Studio Monitor



## NODE LOCATIONS

- 1** Node 1 – Front (Reference)
- 2** Node 2 – Right Wall (Nearfield)
- 3** Node 3 – Right Wall (Upper)
- 4** Node 4 – Back Wall (Upper)



Nodes are positioned to capture spatial acoustic variations within the room.



This setup enabled repeatable, controlled multi-node acoustic testing.

# Flutter Integration and Recovery Testing

Live backend data • partial capture handling • recovery evidence

## 1 Live node status from backend

Online/offline state remains visible while node telemetry is still displayed from the latest capture.

node01

Latest Capture

Estimated SPL   Peak

92.8 dB SPL   46.9 Hz

Details ▾

OFFLINE

node02

Latest Capture

Estimated SPL   Peak

92.9 dB SPL   46.9 Hz

Details ▾

ONLINE

node03

Latest Capture

Estimated SPL   Peak

93.3 dB SPL   46.9 Hz

Details ▾

OFFLINE

node04

Latest Capture

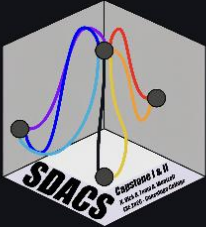
Estimated SPL   Peak

92.7 dB SPL   46.9 Hz

Details ▾

ONLINE

## 2 Capture workflow completes



Capture Complete

Synchronized Room Measurement

Processing Acoustic Results...

Capture complete screen confirms synchronized room measurement

## 3 Result records partial capture

### Capture summary

Capture ID: capture\_20260803T205747074Z  
Collection: Partial  
Completed nodes: 2 of 4 (node02, node04)  
Missing nodes: node01, node03

Completed 2 of 4 nodes; missing nodes are named

### Degraded operation verified

The GUI reports missing nodes and continues showing available results instead of crashing.

## On-slide evidence claim

- ✓ GUI tested against the live Raspberry Pi backend.
- ✓ Offline nodes produced visible status changes, not a crash.
- ✓ Available nodes remained usable during partial capture.
- ✓ Capture summary explicitly reported completed and missing nodes.

## Still collect to complete the slide

- ☐ Backend JSON screenshot
- ☐ API value vs. Flutter card match
- ☐ Backend disconnect / recovery screenshot

Key result: SDACS GUI remains informative during partial node availability and links capture progress to backend result records.



# Backend JSON → Flutter Node Card

*API-versus-Flutter comparison for node04*



Frontend Test Plan Figure 6:  
PowerShell JSON telemetry

```
PS C:\Users\kyle>d -Invoke-RestMethod "http://192.168.5.61:8080/api/nodes/node04/features/latest" | ConvertTo-Json -Depth 10
```

```
{
  "id": 18,
  "timestamp_iso": "99975168",
  "timestamp_us": 99975168,
  "node_id": "node04",
  "topic": "sdacs/node/node04/features",
  "record_type": "features",
  "fw_version": "ota-enable-v2",
  "status": null,
  "capture_state": null,
  "phase": null,
  "seq": 4,
  "rssi": 26624,
  "rms": 4.4E-05,
  "dbfs": -87.2,
  "db_spl": 32.99,
  "f_peak_hz": 46.9,
  "p2p_raw": 2855,
  "zeros": 30,
  "temp_c": 21.08,
  "rh_percent": 62.49,
  "batt_soc_percent": 98.0,
  "batt_voltage_v": 3.0638,
}
```



node04 = node04



32.99 dB = 32.99 dB



46.9 Hz = 46.9 Hz



Firmware version matches



RMS / dBFS / dB SPL /  
f\_peak\_hz match



Battery and environmental  
values match after display  
rounding



Frontend Test Plan Figure 11:  
API-versus-Flutter comparison



node04

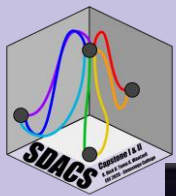
32.99 dB SPL

46.9 Hz peak

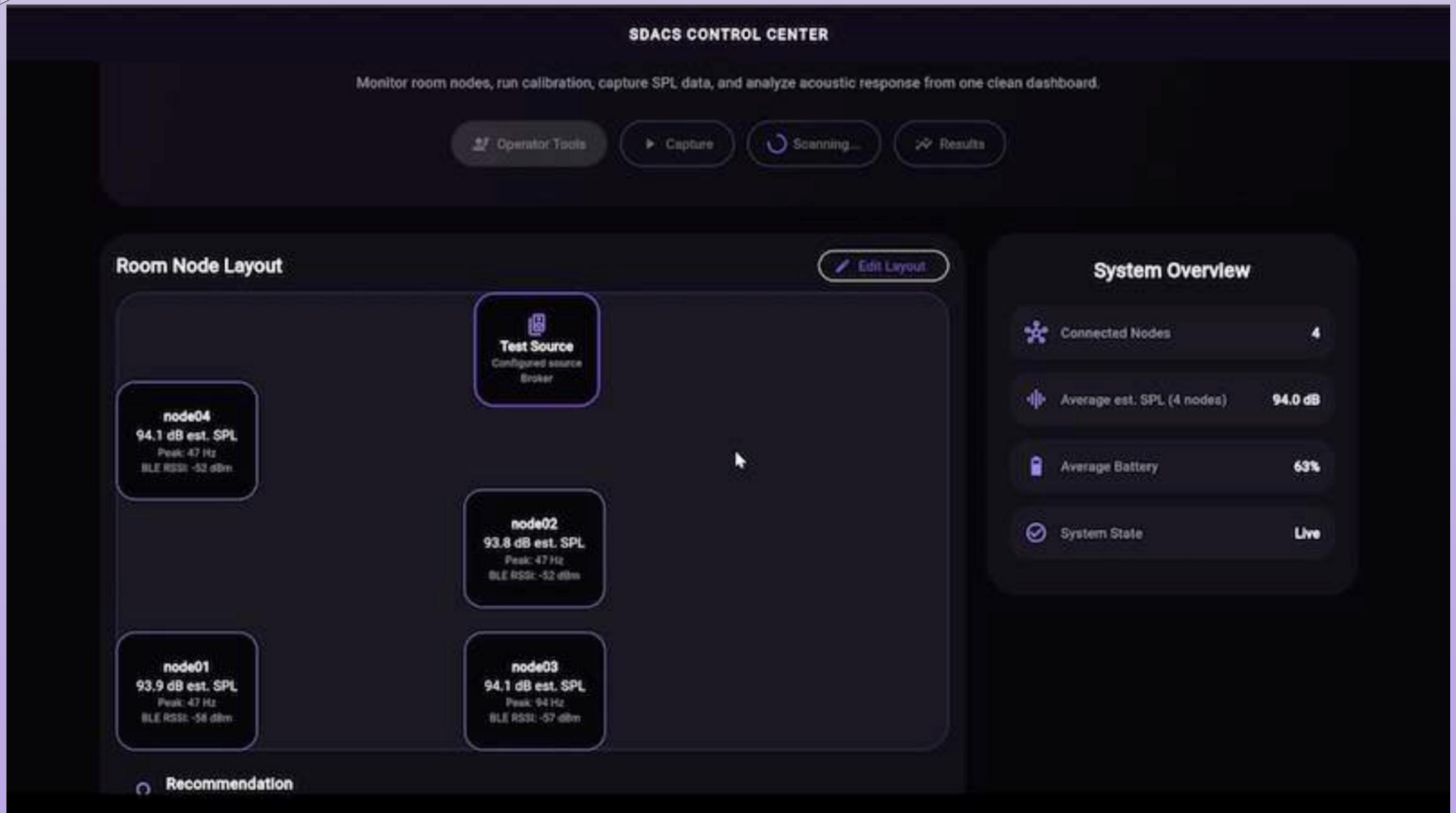
Low - 98% battery



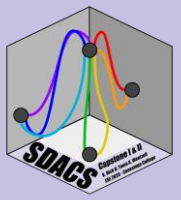
The document records matching node ID, SPL, peak frequency, firmware version, battery, temperature, humidity and RMS values for node04.



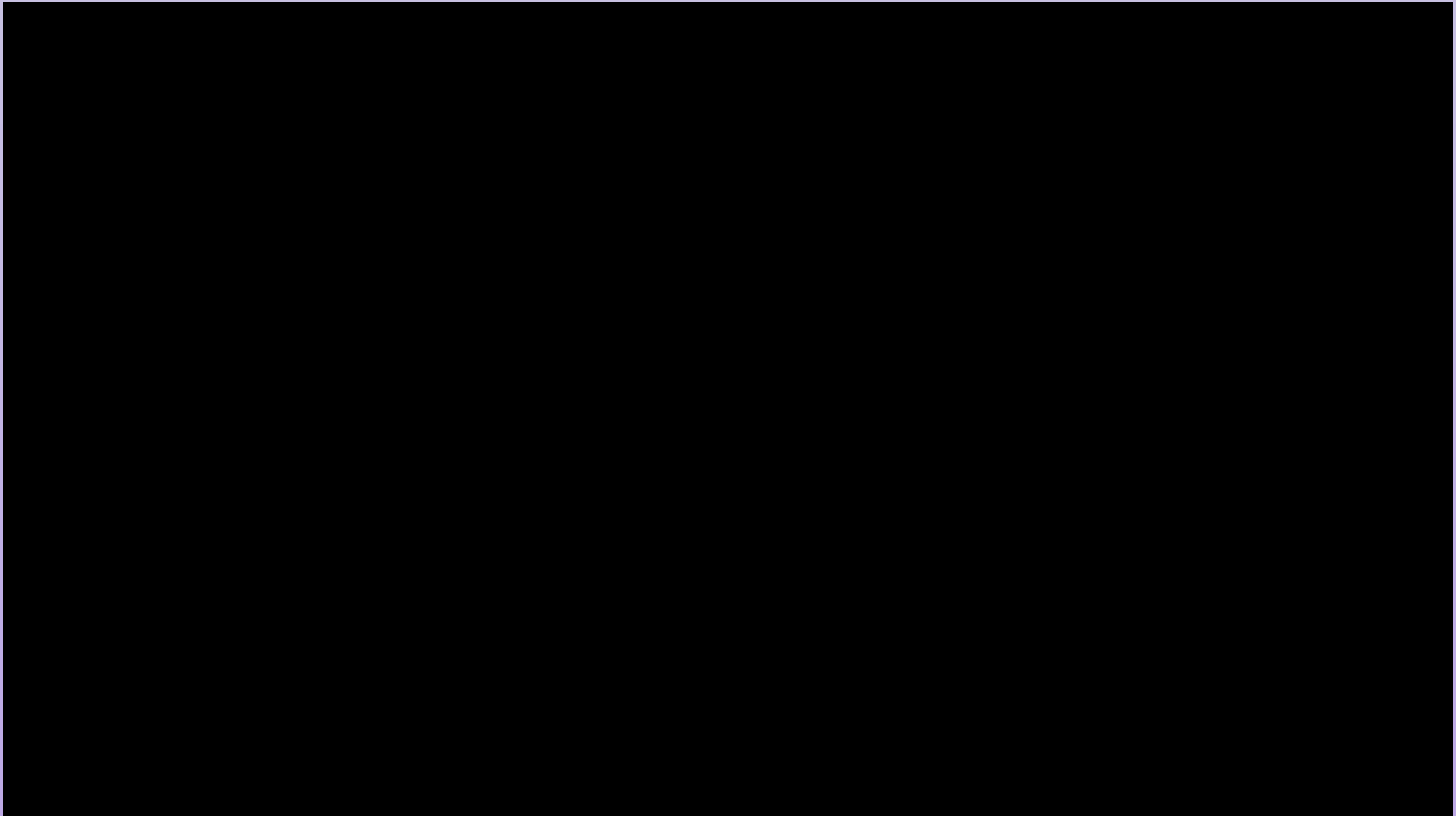
## 9. PROTOTYPING AND TESTING – BLUETOOTH RSSI

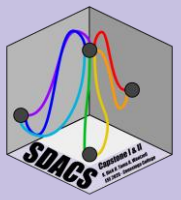




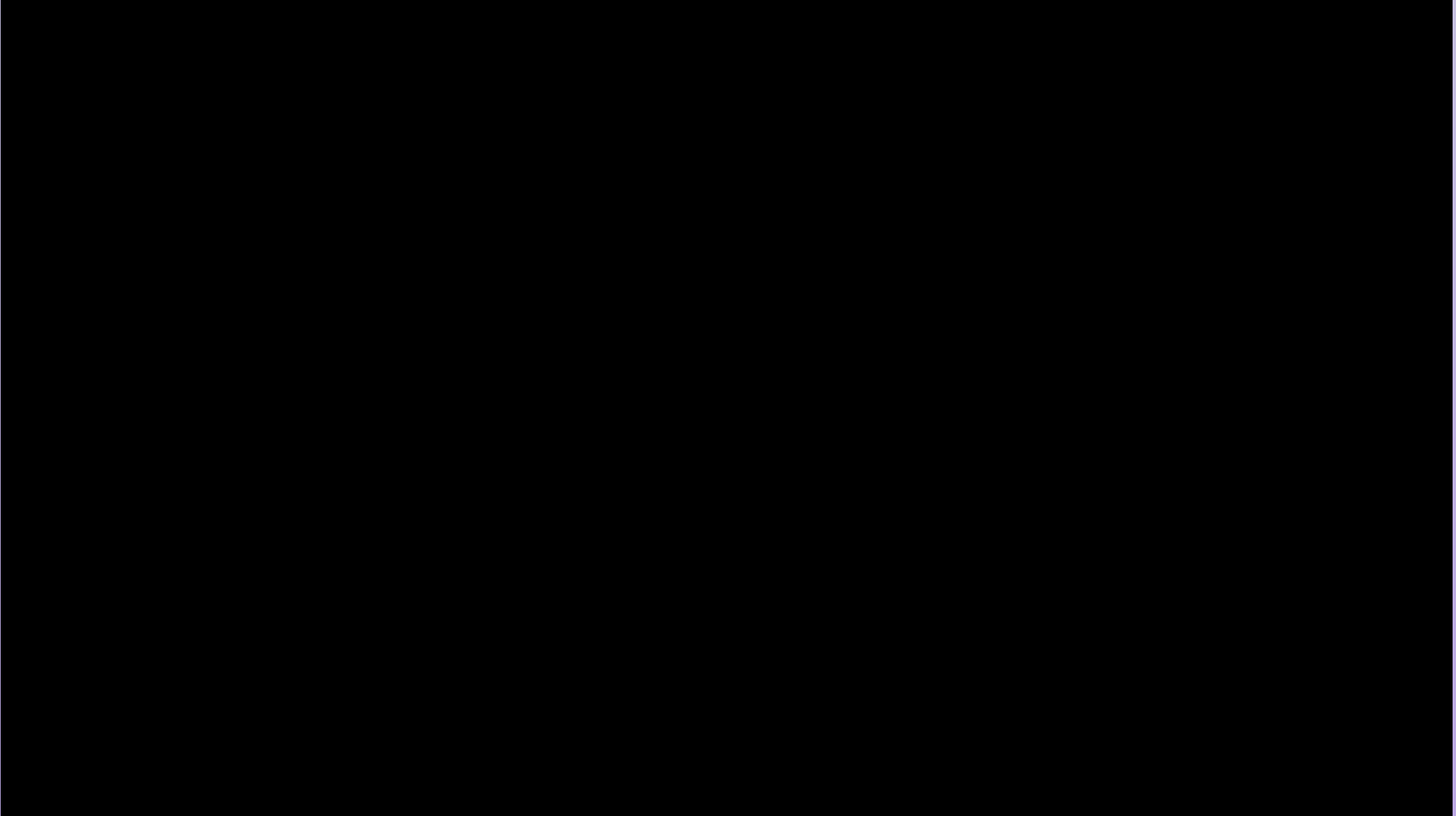


## 5. PROTOTYPING & TESTING – SPL CALIBRATION

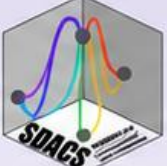




## 5. PROTOTYPING & TESTING – FREQUENCY SWEEP







# 5. PROTOTYPING & TESTING – EDGE IMPULSE

## Representative Early 6-Class Model

### Results

**ACCURACY**  
**58.82%**

### Metrics for Classifier

| METRIC                       | VALUE |
|------------------------------|-------|
| Area Under ROC Curve ⓘ       | 0.58  |
| Weighted average Precision ⓘ | 0.72  |
| Weighted average Recall ⓘ    | 0.71  |
| Weighted average F1 score ⓘ  | 0.71  |

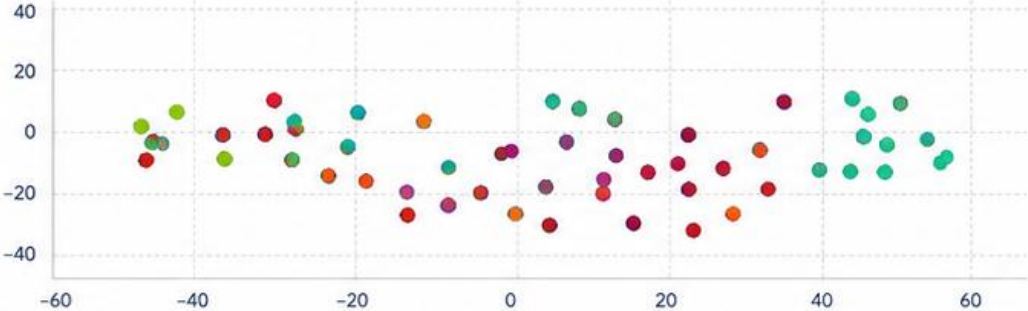
Model version: ⓘ **Unoptimized (Early v1)** ▾

### Confusion Matrix (Simplified) ⓘ

|            |                   | PREDICTED CLASS |                   |            |            |                |           |
|------------|-------------------|-----------------|-------------------|------------|------------|----------------|-----------|
|            |                   | Calib. Tone     | Low Freq. Buildup | Noisy Room | Quiet Room | Speech Present | Uncertain |
| TRUE CLASS | Calib. Tone       | 40%             | 20%               | 5%         | 10%        | 10%            | 15%       |
|            | Low Freq. Buildup | 20%             | 30%               | 10%        | 10%        | 5%             | 25%       |
|            | Noisy Room        | 10%             | 15%               | 35%        | 15%        | 5%             | 20%       |
|            | Quiet Room        | 10%             | 10%               | 15%        | 50%        | 5%             | 10%       |
|            | Speech Present    | 5%              | 10%               | 5%         | 5%         | 40%            | 35%       |
|            | Uncertain         | 5%              | 5%                | 5%         | 5%         | 10%            | 70%       |

### Feature Explorer (t-SNE Projection) ⓘ

- calibration\_tone\_short\_reference – correct
- frequency\_sweep\_20hz\_20khz – correct
- low\_frequency\_buildup – correct
- noisy\_room – correct
- quiet\_room – correct
- speech\_present\_75db – correct
- calibration\_tone\_short\_reference – incorrect
- low\_frequency\_buildup – incorrect
- noisy\_room – incorrect
- speech\_present – incorrect



### Early Model Summary

- 6 classes
- 57 features
- Initial proof of concept



### Observed Challenges

- Substantial class overlap across several labels
- Lower separability for complex or overlapping acoustic scenes



### Engineering Interpretation

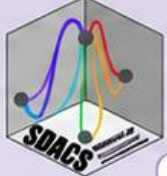
- Motivated later data balancing, feature reduction, and class simplification
- Led to improved separability and stronger model performance



The early multi-class dataset proved feasibility, but performance and class separation were not yet strong enough for a deployment-ready model.



Later optimized model reached **82.4%** validation accuracy.



# 5. PROTOTYPING & TESTING – EDGE IMPULSE

## Model

Model version: ⓘ Quantized (int8) ▼

Last training performance (validation set)

 ACCURACY  
**82.4%**

 LOSS  
**0.42**

Confusion matrix (validation set)

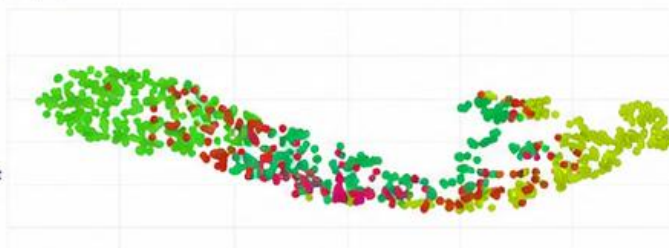
|                        | NOISY | QUIET ROOM WHITE NOISE | SPEECH |
|------------------------|-------|------------------------|--------|
| NOISY                  | 80.8% | 0%                     | 19.2%  |
| QUIET ROOM WHITE NOISE | 0%    | 90.9%                  | 9.1%   |
| SPEECH                 | 15.1% | 9.3%                   | 75.6%  |

Metrics (validation set)

| METRIC                       | VALUE |
|------------------------------|-------|
| Area Under ROC Curve ⓘ       | 0.94  |
| Weighted average Precision ⓘ | 0.83  |
| Weighted average Recall ⓘ    | 0.82  |
| Weighted average F1 score ⓘ  | 0.83  |

Data explorer (classified samples) ⓘ

-  noisy - correct
-  quiet\_room\_white\_noise - correct
-  speech - correct
-  noisy - incorrect
-  quiet\_room\_white\_noise - incorrect
-  speech - incorrect



On-device performance ⓘ

 INFERRING TIME  
**2 ms.**

 PEAK RAM USAGE  
**1.6K**

 FLASH USAGE  
**19.0K**



## Summary Metrics

Validation Accuracy **82.4%**

Validation Loss **0.42**



## Confusion Matrix Summary

- Noisy: 80.8%** correctly classified and **19.2%** confused as Speech.
- Quiet Room White Noise: 90.9%** correctly classified and **9.1%** confused as Speech.
- Speech: 75.6%** correctly classified, **15.1%** confused as Noisy, and **9.3%** confused as Quiet Room White Noise.



## On-Device Performance

Inferencing Time **2 ms.**

Peak RAM Usage **1.6K**

Flash Usage **19.0K**

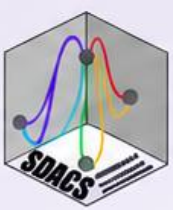


**Final optimized 3-class model:** higher separability, lower feature count, and better suitability for deployment than earlier multi-class versions.



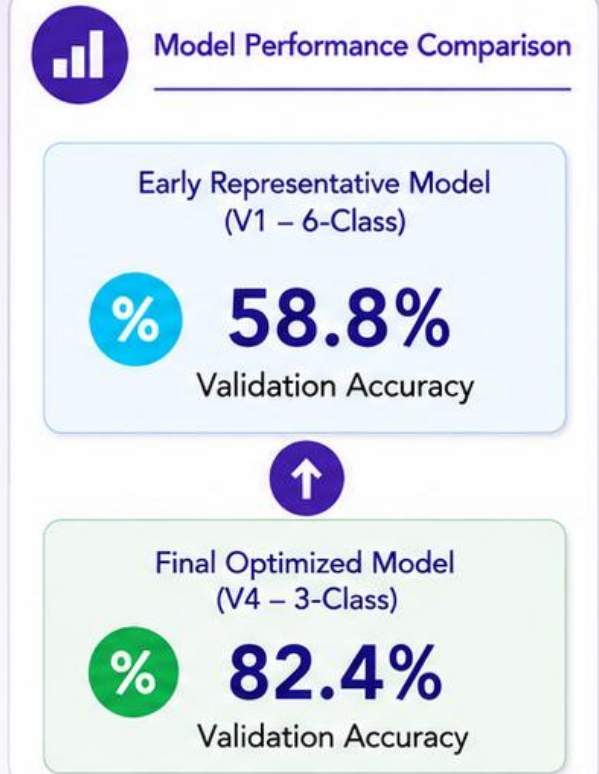
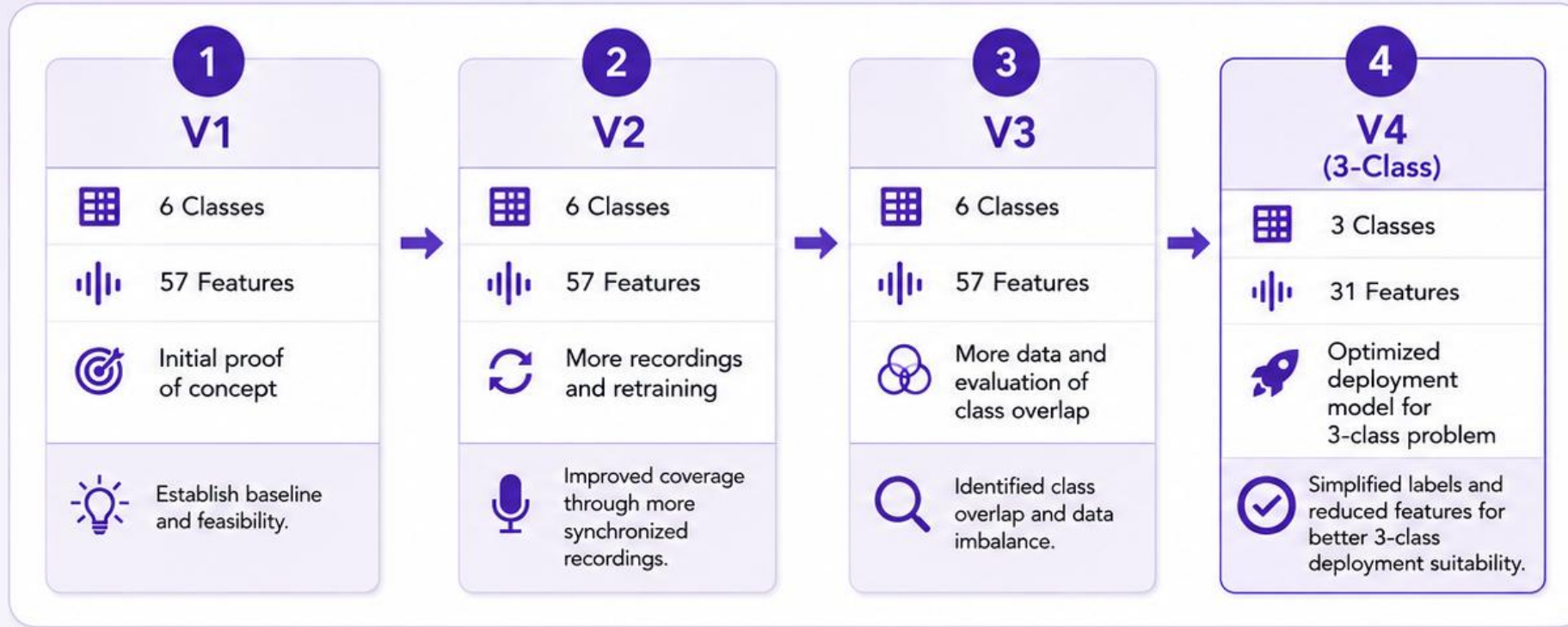
**Improvement over earlier 6-class baseline:**  
**58.8% → 82.4%** validation accuracy.





# 5. PROTOTYPING & TESTING – EDGE IMPULSE

## Model and Dataset Evolution



Engineering Progression



More synchronized recordings



Better dataset balance



Feature reduction  
57 → 31



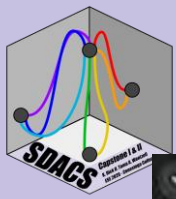
Label simplification  
6 classes → 3 classes



Improved deployment suitability



**Final direction:** a balanced 3-class fused dataset with 31 room-level features provided the best accuracy–deployment trade-off for SDACS.



# 5. PROTOTYPING & TESTING – EDGE IMPULSE

SDACS

localhost56296/#/capture-results

← Capture Results

### Recommendations & Spatial Interpretation

 The room was classified as quiet / white noise.  
The classifier accepted the quiet-room class at 98.6% confidence. This capture can serve as a reference condition for later speech and noisy-room comparisons.

Recommended next action

Save this room layout and capture as the baseline, then keep node positions fixed during comparative tests.

- Investigate any node that remains noticeably louder than the others before treating the room as uniformly quiet.
- Repeat the baseline periodically after firmware, gain, calibration, or room-layout changes.

Evidence used

- Accepted AI class: Quiet Room / White Noise at 98.6% confidence.
- 59 synchronized four-node windows contributed to the capture result.
- node02 was the loudest node; spatial variation was 12.1 dB.
- Dominant acoustic evidence: low-band near 108.3 Hz.

Spatial-map note: use the acoustic profile above to compare relative node levels and frequency-band energy. The map supports room-level interpretation, but it should not be treated as exact source localization.

Backend analysis summary

### Room summary

Nodes used: node01, node02, node03, node04  
Mean estimated SPL: 86.8 dB  
Dominant band: low  
Dominant frequency: 108.3 Hz  
Loudest node: node02  
Quietest node: node04

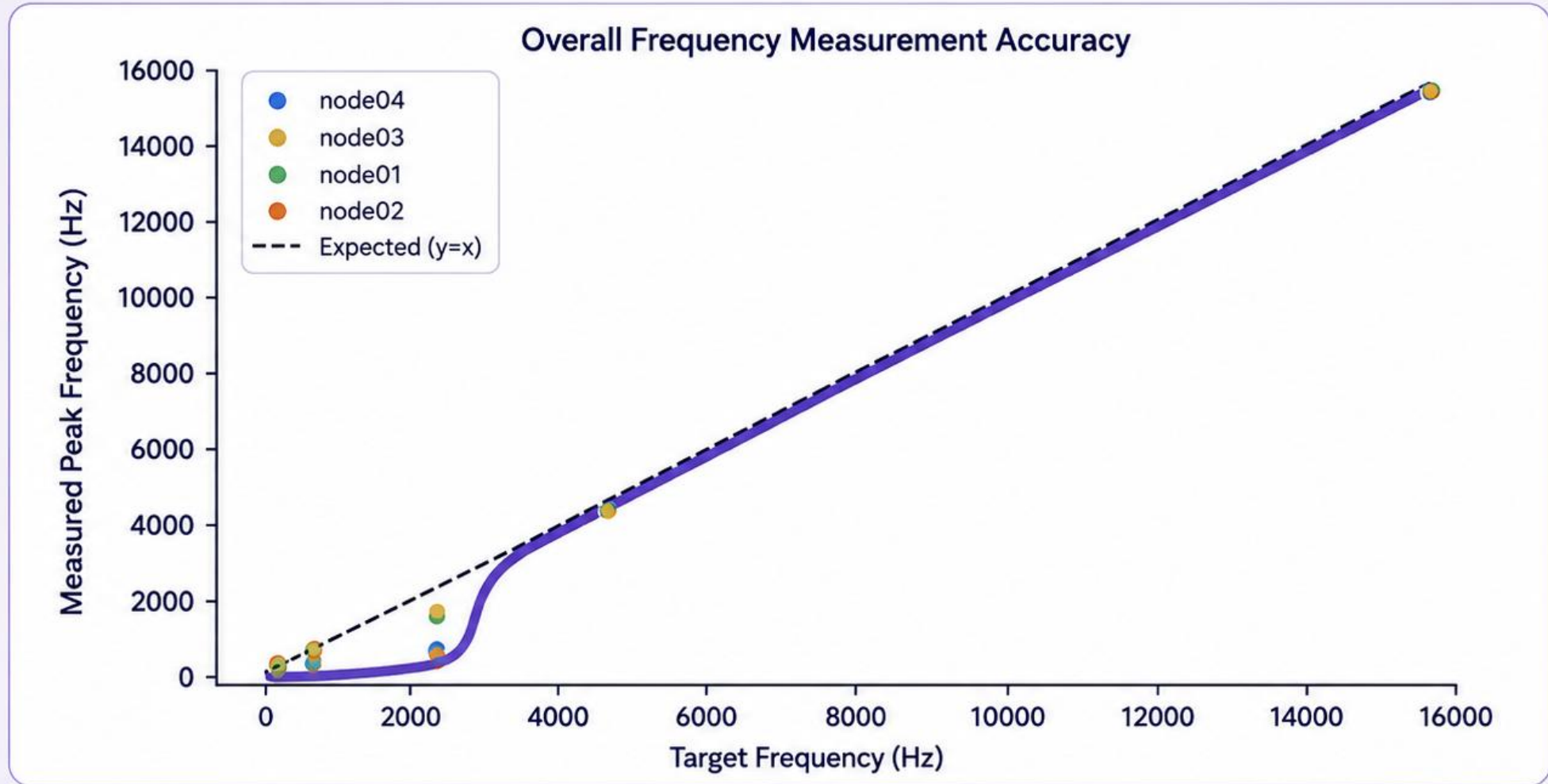
21°

Search

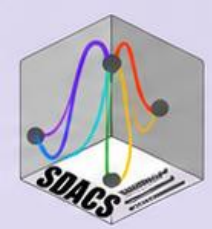
ENG US 11:38 AM 2025-08-07



# 6. RESULTS & ANALYSIS – HARDWARE (NFDA)



**KEY TAKEAWAY:** All nodes track the expected frequency ( $y=x$ ) closely above the low-frequency roll-off region. Deviation is primarily observed at lower frequencies.



# 6. RESULTS & ANALYSIS - FIRMWARE

*Final SDACS dataset prepared for Edge Impulse*



## Dataset Snapshot

- 33** capture attempts
- 31** complete four-node sessions
- 30** balanced sessions selected
- 1,770** balanced fused records
- 31** aggregate input features

## Balanced 3-Class Distribution



### Quiet Room

10 sessions  
590 rows



### Speech

10 sessions  
590 rows



### Noisy

10 sessions  
590 rows



## Recommended Edge Impulse Split

- Training.csv**  
24 sessions total, 1,416 rows  
8 sessions per class
- Testing.csv**  
6 sessions total, 354 rows  
2 sessions per class
- Validation\_reference.csv**  
3 sessions total, 177 rows  
1 session per class
- Training\_core.csv**  
21 sessions total, 1,239 rows  
7 sessions per class



## Data Quality

- ✓ 2 incomplete sessions excluded
- ✓ Latest capture: capture\_20260804173354
- ✓ Latest capture is a complete noisy session with 59 rows per node, 236 total rows
- ✓ Latest capture reserved in testing split
- ✓ Row validity passed
- ✓ Sample-rate checks passed
- ✓ Scene-metric checks passed
- ✓ Clipping checks passed

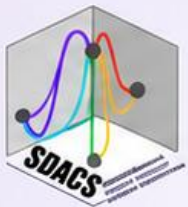


**Baseline sanity check: Random Forest achieved 91.5% accuracy on the strict held-out split.**



**Fusion method:** 31 permutation-invariant room-level features derived from synchronized 4-node captures.





# 7. CONCLUSION



## SDACS is...



Distributed acoustic calibration network



Simultaneous (group) node control



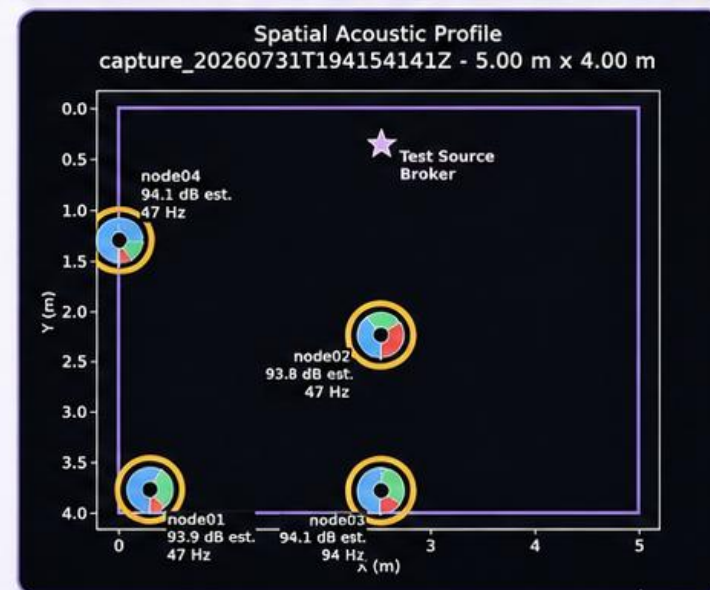
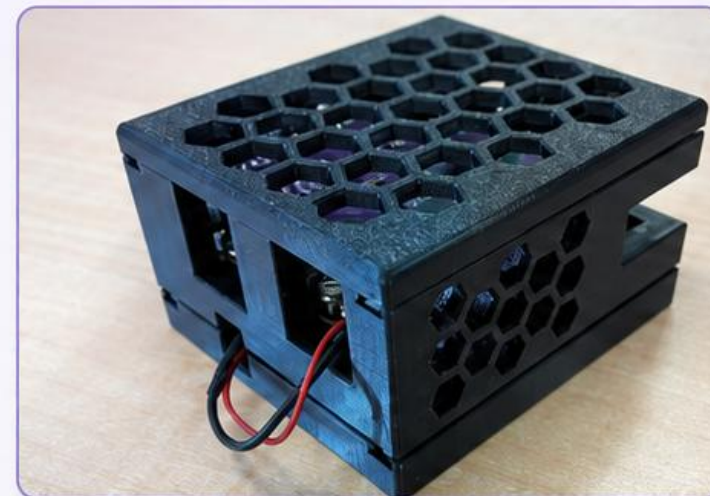
Wireless data acquisition



Frequency analysis and room treatment



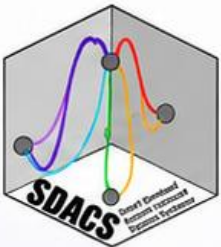
**A working prototype, room to expand and improve**



## KEY TAKEAWAY:

SDACS demonstrates a working distributed sensing prototype with synchronized capture, wireless telemetry, and strong room-analysis potential.





# 8. RECOMMENDATIONS



## 1 Expand the network

- 8+ nodes for higher-accuracy spatial profiles



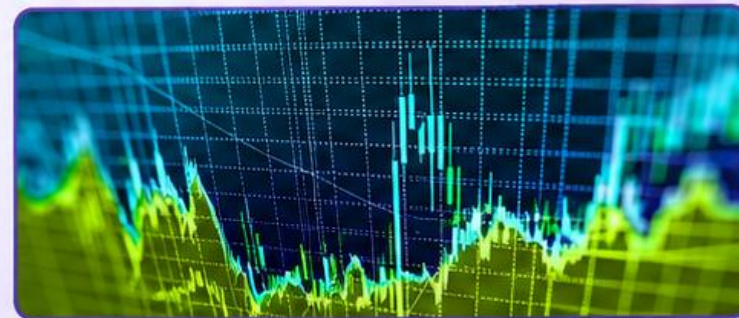
## 2 Detailed FFT analysis

- Instrument profile recommendations
- Harmonic identification



## 3 Minimize / improve hardware

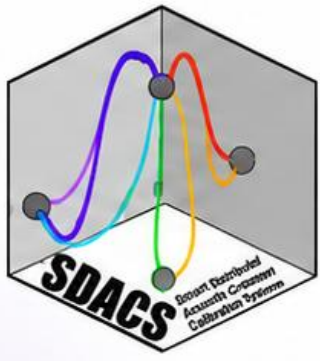
- SoC integration (combine hardware on one board)
- Lobe mic
- Smaller product (reduce by 50%)



### KEY TAKEAWAY

Future work should improve spatial resolution, spectral intelligence, and hardware integration for a more accurate and compact SDACS platform.





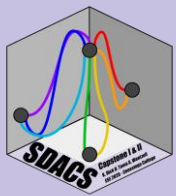
# 10. Q & A

Questions and Discussion

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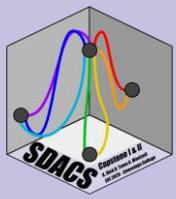
— Thank you for your attention. —



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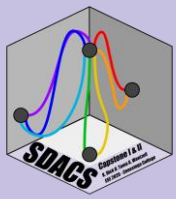
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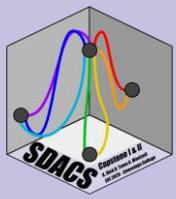
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